Title: Diabetes Prediction Using BRFSS Dataset

Abstract. This paper evaluates diabetes prediction using BRFSS survey records with logistic regression, random forest, and feature selection. The study uses health indicators including BMI, age, high blood pressure, cholesterol, physical activity, and general health.

Methods. We preprocess missing survey responses, balance classes, select informative features with BRFSS risk factors, and compare models using train test splits.

Results. Random forest reports higher F1 and AUC than logistic regression. Feature importance highlights BMI, age, high blood pressure, and general health as strong predictors. Figure 1: Model comparison across F1 and AUC. Table 1: Diabetes prediction metrics.

Limitations. BRFSS is self reported survey data, so reporting bias and cross sectional design limit causal interpretation. Future work should validate on clinical records and external cohorts.